7 An induced nuclear fission reaction may be represented by the equation

$$^{235}_{92}U + ^{1}_{0}n \longrightarrow ^{141}_{56}Ba + ^{92}_{36}Kr + 3^{1}_{0}n$$

The variation with nucleon number A of the binding energy per nucleon B_E is illustrated in Fig. 7.1.

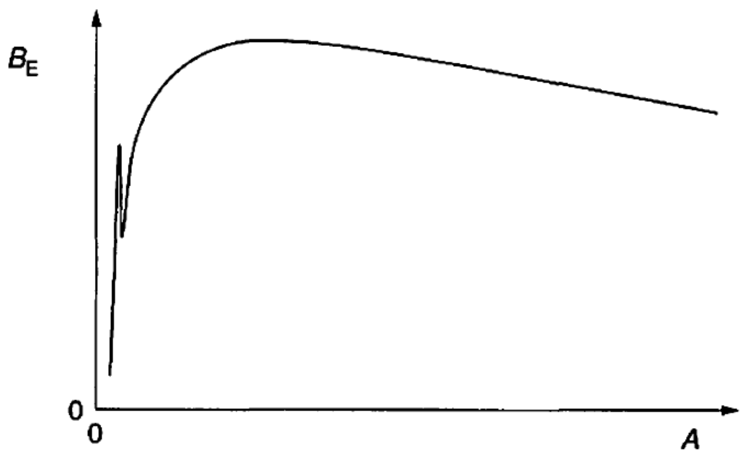


Fig. 7.1

(a) State an approximate value, in MeV, for the maximum binding energy per nucleon.

maximum binding energy per nucleon = …………………… MeV [1]

L1 Maximum binding energy per nucleon = **8.8 MeV** **(or 9 MeV)** (for Fe-56) **1**

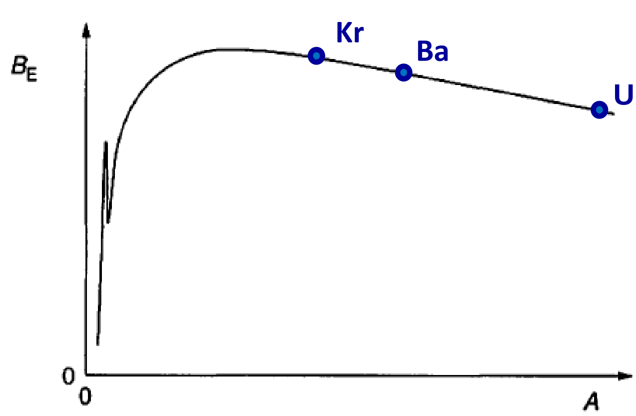
(b) On Fig. 7.1, mark approximate positions for the nuclei of

(i) uranium-235 (label the position U),

(ii) barium-141 (label the position Ba),

(iii) Krypton-92 (label the position Kr). [3]

Solution:



Mark scheme:
1 mark – In the ascending order, Kr, Ba and U **1**
1 mark – All markings on the right-hand side of the peak (A = 56) **1**
1 mark – Relative positions **1**

	(c)	By reference to binding energy per nucleon, explain why energy is released in this fission reaction.	
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		In the fission reaction, the **product** nuclides (Kr and Ba) have a **greater binding energy per nucleon** than the reactants (U and n, where n has zero binding energy). **Since the total number of nucleons is unchanged in the reaction**, the **total binding energy** of the products is greater than that of the reactants. This means that the **total energy released in the formation of Kr and Ba** is **more than** the **total energy absorbed during the disintegration of U**. Hence there is a *net* release in energy.	**1** **1** **1**

8	Read the passage below and answer the questions that follow